

Chapter 1

Introduction and review

1.1 Course overview

1.1.1 Instructor

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Research Interests:

- Structural Engineering, Performance-Based Engineering
- Bluff-Body Aerodynamics
- Reduced-Order Modeling
- Natural Hazard Modeling (Hurricane, Earthquake, Wildfire)
- Deep Learning and Computer Vision

1.1.2 Course modules

This course will be comprised of 4 modules:

- Introduction and Review:
- Single Component: Slope-Deflection Method
- Frame Structure: Direct Stiffness Method
- Second-Order Effects and Nonlinear Analysis

Coding for practical application will be integrated throughout the course, with a focus on programming techniques relevant to structural analysis. **Python** will be the primary programming language used in this course.

1.2 Fundamental concepts

1.2.1 Units in structural engineering

Both metric units (i.e., SI units) and imperial units will be used in this course. Please make sure you are familiar with the conversion between both systems.

- Many countries around the world began adopting the metric system in the late 19th century.
- In the U.S., imperial units are still in use. However, U.S. structural engineers are required to use the metric system for most international projects.

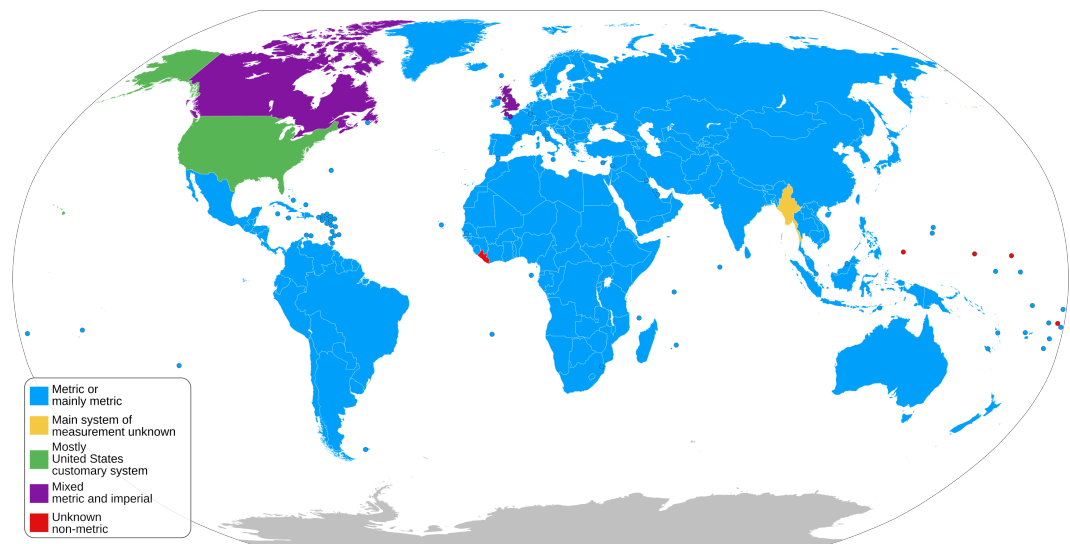


Figure 1.1: Countries using the metric (SI), imperial and US customary systems as of 2019 (Wikipedia)

Unit conversion

On September 23, 1999 NASA lost the \$125 million Mars Climate Orbiter spacecraft after a 286-day journey to Mars. Thrusters used to help point the spacecraft had, over the course of months, been fired incorrectly because data used to control the wheels were calculated in incorrect units. **Lockheed Martin, which was performing the calculations, was sending thruster data in English units (pounds) to NASA, while NASA's navigation team was expecting metric units (Newtons).** (<http://spacemath.gsfc.nasa.gov/slashweekly/6Page53.pdf>)

4a. Lengths, Areas and Volumes - English Units to S.I. Units and Back

<i>To Convert From</i>	<i>multiply</i>	<i>to get / To Convert from</i>	<i>multiply by</i>	<i>to get</i>
Length				
inches, in	<u>25.4</u>	millimeters, mm	0.0393701	inches, in
inches, in	<u>0.0254</u>	meters, m	39.3701	inches, in
feet, ft	<u>0.3048</u>	meters, m	3.280840	feet, ft
yards, yd	<u>0.9144</u>	meters, m	1.09361	yards, yd
miles, mi	<u>1.609344</u>	kilometers, km	0.621371	miles, MI
Area				
square inch, in ² , si	<u>645.16</u>	square millimeters, mm ²	0.00155000	square inch, in ²
square inch, in ² , si	<u>0.00064516</u>	square meter, m ²	1550.00	square inch, in ²
square feet, ft ² , sf	<u>0.09290304</u>	square meter, m ²	10.7639	square feet, ft ² , sf
square yard, yd ² , sy	<u>0.836127</u>	square meter, m ²	1.19599	square yard, yd ²
Volume				
cubic inch, in ³ , ci	<u>16387.064</u>	cubic millimeters, mm ³	61.0237 x 10 ⁻⁶	cubic inch, in ³
cubic inch, in ³ , ci	<u>16.387064</u>	cubic centimeters, cc, cm ³	0.06102374	cubic inch, in ³
cubic foot, ft ³ , cf	<u>0.0283168</u>	cubic meters, m ³	35.3147	cubic foot, ft ³ , cf
cubic yard, yd ³ , cy	<u>0.764555</u>	cubic meters, m ³	1.30795	cubic yard, yd ³
gallon, gal.	<u>3.78541</u>	Liters, l = 0.001 m ³	0.264172	gallon, gal.
Other Geometric Properties of a Cross-section				
Moment of Inertia, in ⁴	<u>416231</u>	mm ⁴ = 10 ⁻¹² m ⁴	2.402 x 10 ⁻⁶	in ⁴
Section Modulus, in ³	<u>16387.064</u>	mm ³ = 10 ⁻⁹ m ³	61.0237 x 10 ⁻⁶	in ³

Figure 1.2: Unit conversion 1 (ah-engr.com)

4b. Force, Stress, Loads and Mass - English Units to S.I. Units and Back

<i>To Convert From</i>	<i>multiply</i>	<i>to get / To Convert from</i>	<i>multiply by</i>	<i>to get</i>
Force				
pound, lb	4.44822	newton, N	0.224809	lb
kilopound, kip = 1000 lb	4.44822	kilonewton, kN = 1000 N	0.224809	kip = 1000 lb
Stress [Force per Unit Area]				
pounds per sq. ft, lb/ft ² , psf	47.8803	pascals, Pa = N/m ²	0.0208854	lb/ft ² , psf
psf	0.0478803	kilopascal, kPa, kN/m ²	20.8854	psf
kilopounds per sq. ft, ksf	47.8803	kPa	0.0208854	ksf
pounds per sq. inch, psi	6894.76	Pa	0.000145038	psi
psi	6.89476	kPa	0.145038	psi
kilopounds per sq. in, ksi	6.89476	megapascal, MPa = MN/m ² = N/mm ²	0.145038	ksi
Bending Moment, Torque [Force x Length]				
pound-foot, lb-ft	1.35582	newton-meter, N-m	0.737561	lb-ft
pound-inch, lb-in	0.112985	N-m	8.850732	lb-in
Distributed (or "Running") Load [Force per Unit Length]				
pounds per (linear) foot, plf	14.5939	newtons per meter, N/m	0.06852178	plf
pounds per (linear) inch, pli	175.127	N/m	0.00571015	pli

Figure 1.3: Unit conversion 2 (ah-engr.com)

1.2.2 Structural Performance

Key concepts in structural performance:

- **Strength** – ability to resist loads without failure
- **Stiffness** – ability to resist deformation
- **Stability** – ability to remain in equilibrium
- **Serviceability** – ability to function without discomfort

1.2.3 Idealization

Idealization is a process in which an actual structure and the loads acting on it are replaced by simpler models for the purpose of analysis.

Structure

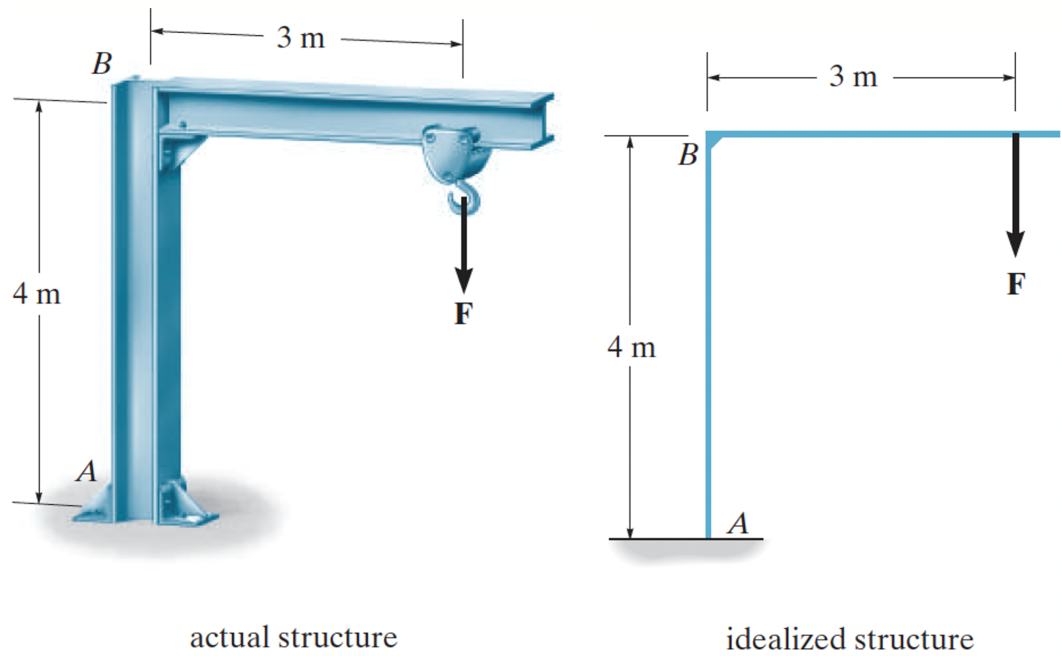


Figure 1.4: Example structure

Load

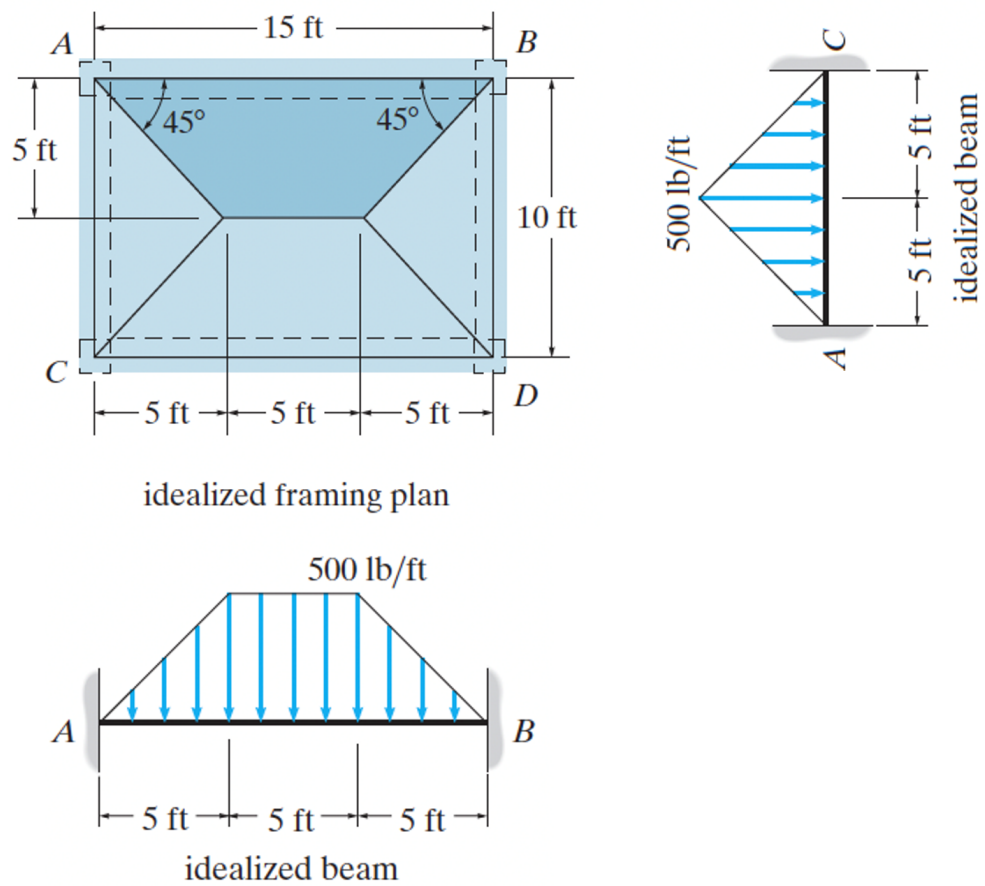


Figure 1.5: Example load

Supports

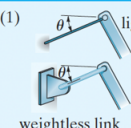
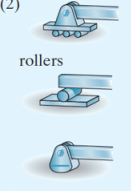
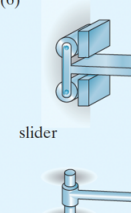
TABLE 2.1 Supports for Coplanar Structures			
Type of Connection	Idealized Symbol	Reaction	Number of Unknowns
(1)  light cable weightless link			One unknown. The reaction is a force that acts in the direction of the cable or link.
(2)  rollers rocker			One unknown. The reaction is a force that acts perpendicular to the surface at the point of contact. This support is assumed to be able to exert a force in either direction.
(3)  smooth contacting surface			One unknown. The reaction is a force that acts perpendicular to the surface at the point of contact.
(4)  smooth pin-connected collar			One unknown. The reaction is a force that acts perpendicular to the surface at the point of contact.
(5)  smooth pin or hinge			Two unknowns. The reactions are two force components.
(6)  slider fixed-connected collar			Two unknowns. The reactions are a force and a moment.
(7)  fixed support			Three unknowns. The reactions are the moment and the two force components.

Figure 1.6: Supports

1.2.4 Free-body diagram

Newton's first law (law of inertia)

A body remains at rest, or continues to move with constant velocity, unless acted upon by a nonzero resultant force. Therefore, every member and every complete structure that is at rest must satisfy

$$\sum \mathbf{F} = \mathbf{0}.$$

Both sides of this equation are vectors. In structural analysis, equilibrium is written as scalar force and moment equations:

Two-dimensional equilibrium

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M_z = 0$$

Three-dimensional equilibrium

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum F_z = 0$$

$$\sum M_x = 0$$

$$\sum M_y = 0$$

$$\sum M_z = 0$$

Constructing a free-body diagram

A free-body diagram isolates the body being analyzed and replaces every connection to its surroundings with the corresponding reaction forces and moments.

1. Select and isolate the entire structure or one member.
2. Show all applied loads and replace each support or connection by its possible reactions.
3. Define the coordinate axes and a positive moment convention.
4. Write one independent equilibrium equation for each available direction of translation or rotation.

Example 1: statically determinate beam

Consider a simply supported beam of length L carrying a concentrated load P at midspan. The pin at A provides two reaction components, and the roller at B provides one.

With counterclockwise moments positive,

$$\begin{aligned}\sum F_x = 0 : \quad & F_{Ax} = 0, \\ \sum F_y = 0 : \quad & F_{Ay} + F_{By} - P = 0, \\ \sum M_A = 0 : \quad & -P\frac{L}{2} + F_{By}L = 0.\end{aligned}$$

There are three independent equilibrium equations and three unknown reactions, so the solution is unique:

$$\boxed{F_{Ax} = 0, \quad F_{Ay} = \frac{P}{2}, \quad F_{By} = \frac{P}{2}}.$$

This beam is **statically determinate**: all reactions can be obtained from equilibrium alone.

Example 2: statically indeterminate beam

Replace the pin at A with a fixed support. The free-body diagram now contains four unknown reactions: F_{Ax} , F_{Ay} , M_A , and F_{By} .

Equilibrium gives

$$\begin{aligned} F_{Ax} &= 0, \\ F_{Ay} + F_{By} - P &= 0, \\ M_A - P\frac{L}{2} + F_{By}L &= 0. \end{aligned}$$

There are only three equilibrium equations for four unknown reactions. In terms of the undetermined reaction moment M_A ,

$$F_{Ax} = 0, \quad F_{Ay} = \frac{P}{2} + \frac{M_A}{L}, \quad F_{By} = \frac{P}{2} - \frac{M_A}{L}.$$

For example, $M_A = 0$, $M_A = PL$, and $M_A = -PL$ each produce a different set of reactions that satisfies equilibrium. The actual structure has only one response; an additional **deformation-compatibility equation**, together with a force–deformation relation, is required to find it. This beam is **statically indeterminate to the first degree**.

1.3 Determinacy and stability

1.3.1 Determinacy

Statically Determinate Structures: The equilibrium equations provides both the *necessary* and *sufficient* conditions for equilibrium. When all the forces can be strictly obtained from these equations, the structure is termed as statically determinate.

Statically Indeterminate Structures: Structures having more unknown forces than available equilibrium equations. These structures require additional compatibility conditions for analysis and may have multiple solutions.

Degree of indeterminacy

For 2D analysis, there are at most three equilibrium equations for each part, so if there is a total of n components and r constraints (i.e., force and moment reactions), we have

- $r = 3n$, statically determinate
- $r > 3n$, statically indeterminate

If a structure is statically indeterminate, additional equations (compatibility equations) are needed to solve for the unknowns. These compatibility conditions relate the applied loads and reactions to the displacement or slope at different points in the structure.

The number of compatibility equations required must be equal to the **degree of indeterminacy** $r - 3n$ of the structure.

1.3.2 Stability

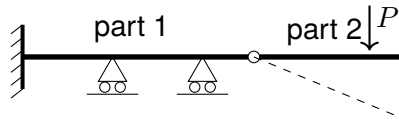
For a structure (or its members) to maintain equilibrium, it is not only necessary to satisfy the equations of equilibrium, but the members must also be **properly constrained** by their supports. There are two situations where the conditions for **proper constraint** are not met (unstable):

1. Partial Constraints. Instability arises when a structure or member has **fewer reaction components than equilibrium equations**, $r < 3n$. In such cases, the structure is only partially constrained.

2. Improper Constraints. Even if the number of reactions equals the number of equilibrium equations, instability can still occur when the **support arrangement is improper**. Two typical cases are:

- *Concurrent reactions:* If all reaction forces are concurrent at a point, rotation may occur.
- *Parallel reactions:* If all reaction forces are parallel, instability occurs under certain loadings.

3. Local Instability: Collapsible Mechanisms. Even when the global count passes and no obviously improper support arrangement is present, a structure can still hide a mechanism. Hibbeler classifies this as *local instability*: even with $r \geq 3n$, a structure is unstable if “some of the components form a collapsible mechanism” (Hibbeler & Tan, 10th ed., Eq. 2–4). The rule $r \geq 3n$ is a *necessary* condition, not a *sufficient* one: $r - 3n$ equals the degree of indeterminacy *only if the structure is properly constrained*. A surplus of restraints in one part can numerically “pay for” a missing restraint in another part, so the partial constraint of one part is hidden by the global count.



Example. Part 1 carries a fixed support and two rollers; part 2 hangs from an internal hinge with no support of its own. Global count: $n = 2$ parts $\Rightarrow 3n = 6$ equations; unknowns = 5 reactions $(3+1+1)+2$ hinge forces = 7. Since $7 > 6$, the count declares the beam “statically indeterminate, degree 1.” But the hinge transmits no moment: part 2 alone has only 2 unknowns for 3 equilibrium equations, so moment equilibrium of part 2 about the hinge cannot be satisfied under the load P — part 2 simply rotates (dashed line). The two *surplus* restraints under part 1 masked the *missing* restraint of part 2: the structure is a **mechanism**, not an indeterminate system.

How to detect local instability:

- Read $r \geq 3n$ only as a screening result; it never certifies stability by itself.
- **Check each member or component.**
- **Treat internally stable portions as single rigid bodies.**
- **Solve the equilibrium equations.**

1.3.3 Why indeterminate structures?

Advantages:

- Redundancy provides alternate load paths
- Smaller deflections
- Better moment distribution

Disadvantages:

- More complex analysis required
- Sensitive to support settlement
- Temperature effects cause stresses
- Fabrication errors can cause stresses

1.4 Superposition and beam analysis

1.4.1 Principle of superposition

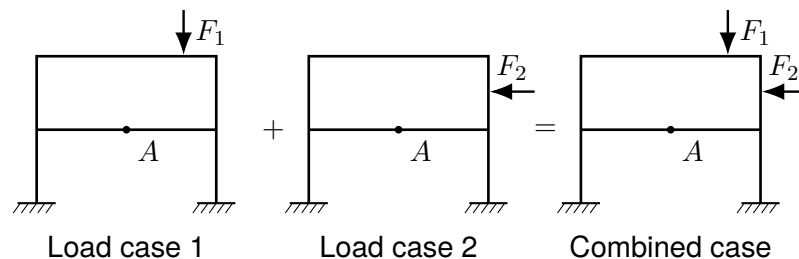
The principle of superposition states that the response caused by several loads acting together equals the sum of the responses caused by each load acting separately. It applies when the structural model is *linear*.

Assumptions

- **Linear-elastic material behavior:** stress is proportional to strain, so unloading follows the same path and no permanent deformation remains.
- **Small deformation:** displacements and rotations are small enough that equilibrium can be written using the original, undeformed geometry.

Applications

The diagrams below are not three different structures. They are three load cases for the *same* structure: load F_1 alone, load F_2 alone, and both loads acting together.



Let $R(F)$ denote any response produced by load F , such as a displacement, support reaction, shear force, or bending moment. For the two load cases,

$$R(F_1 + F_2) = R(F_1) + R(F_2).$$

Additivity and proportionality

Superposition includes two related properties. For load cases F_1 and F_2 and constants a and b ,

$$R(aF_1 + bF_2) = aR(F_1) + bR(F_2).$$

This equation expresses **additivity** (responses from separate load cases can be added) and **proportionality** (scaling a load scales its response by the same factor).

At point A in the preceding structure,

$$u_A(F_1 + F_2) = u_A(F_1) + u_A(F_2),$$

$$v_A(F_1 + F_2) = v_A(F_1) + v_A(F_2),$$

$$V(x; F_1 + F_2) = V(x; F_1) + V(x; F_2),$$

$$M(x; F_1 + F_2) = M(x; F_1) + M(x; F_2).$$

Here, u_A and v_A are displacements at point A ; $V(x)$ and $M(x)$ are shear and moment at section x . Each response must be evaluated at the same location in every load case.

When superposition does not apply

If the response is nonlinear, or if the structural model changes, then

$$R(F_1 + F_2) \neq R(F_1) + R(F_2).$$

- **Material nonlinearity:** yielding, cracking, or another nonlinear stress–strain response.
- **Geometric nonlinearity:** the deformed geometry affects equilibrium, as in P – Δ analysis.
- **Changing model:** contact, supports, connections, or load paths change state.

1.4.2 Beam analysis

Euler-Bernoulli Beam Theory

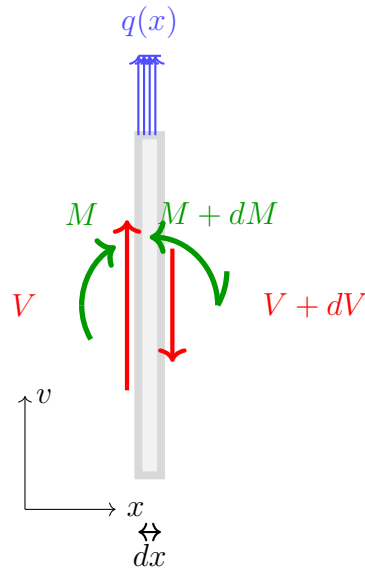


Figure 1.7: Free-body diagram of a beam element

The Euler-Bernoulli beam theory provides the fundamental framework for analyzing the bending behavior of beams under transverse loads. The theory assumes that plane sections remain plane and perpendicular to the neutral axis after deformation.

The governing differential equation for beam deflection is:

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 v}{dx^2} \right) = q(x) \quad (1.1)$$

where:

- E = Young's modulus (elastic modulus)
- I = second moment of area (moment of inertia)
- $v(x)$ = vertical displacement (deflection)
- $q(x)$ = distributed load per unit length
- x = position along the beam axis

For beams with constant flexural rigidity ($EI = \text{constant}$), this simplifies to:

$$EI \frac{d^4 v}{dx^4} = q(x) \quad (1.2)$$

Relationships Between Internal Forces and Displacement

The key relationships in beam theory connect the internal forces to the derivatives of the vertical displacement:

Moment-Curvature Relationship:

$$M(x) = EI \frac{d^2v}{dx^2} \quad (1.3)$$

This shows that the bending moment equals the flexural rigidity times the second derivative of vertical displacement (curvature).

Shear Force-Displacement Relationship:

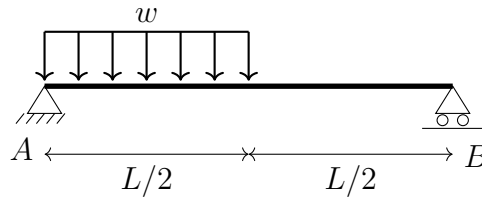
$$V(x) = \frac{dM}{dx} = EI \frac{d^3v}{dx^3} \quad (1.4)$$

The shear force equals the first derivative of the moment, or equivalently, the third derivative of vertical displacement times EI .

Load-Displacement Relationship:

$$q(x) = \frac{dV}{dx} = EI \frac{d^4v}{dx^4} \quad (1.5)$$

These equations form a complete system for solving beam deflection problems. Given appropriate boundary conditions, we can integrate to find the displacement function $v(x)$ and subsequently determine all internal forces.

Example 1: simply supported beam with a uniform load on the left half

Free-body diagram of member AB. Equilibrium of the whole member gives the reactions:

$$\sum M_B = 0 : \quad R_A = \frac{3wL}{8}, \quad \sum F_y = 0 : \quad R_B = \frac{wL}{8}.$$

Governing equations by region. With the load on the left half only, the beam is treated as two regions with separate integration constants ($C_{1L}, \dots, C_{4L}$ and $C_{1R}, \dots, C_{4R}$):

$$EIv_L''''(x) = -w \quad \left(0 \leq x \leq \frac{L}{2}\right), \quad EIv_R''''(x) = 0 \quad \left(\frac{L}{2} \leq x \leq L\right).$$

Integrating once per step (shear, moment, slope, deflection):

$$\begin{array}{ll} V(x) : & EIv_L'''(x) = -wx + C_{1L} & EIv_R'''(x) = C_{1R} \\ M(x) : & EIv_L''(x) = -\frac{w}{2}x^2 + C_{1L}x + C_{2L} & EIv_R''(x) = C_{1R}x + C_{2R} \end{array}$$

Note that $C_{1L} = V_L(0) = R_A$: the first integration constant of the loaded region is the left reaction.

Continuing the integration (slope, then vertical deflection):

$$\text{slope: } EIv'_L(x) = -\frac{w}{6}x^3 + \frac{C_{1L}}{2}x^2 + C_{2L}x + C_{3L} \quad EIv'_R(x) = \frac{C_{1R}}{2}x^2 + C_{2R}x + C_{3R}$$

$$\text{deflection: } EIv_L(x) = -\frac{w}{24}x^4 + \frac{C_{1L}}{6}x^3 + \frac{C_{2L}}{2}x^2 + C_{3L}x + C_{4L}$$

$$EIv_R(x) = \frac{C_{1R}}{6}x^3 + \frac{C_{2R}}{2}x^2 + C_{3R}x + C_{4R}$$

8 unknowns $\Rightarrow$ 8 equations. Boundary conditions (displacement and moment at each support):

$$\textcircled{1} v_L(0) = 0 \quad \textcircled{2} EIv''_L(0) = 0 \quad \textcircled{3} v_R(L) = 0 \quad \textcircled{4} EIv''_R(L) = 0$$

Continuity conditions at $x = L/2$ (deflection, slope, moment, shear – no point load at the junction):

$$\textcircled{5} v_L\left(\frac{L}{2}\right) = v_R\left(\frac{L}{2}\right)$$

$$\textcircled{6} v'_L\left(\frac{L}{2}\right) = v'_R\left(\frac{L}{2}\right)$$

$$\textcircled{7} EIv''_L\left(\frac{L}{2}\right) = EIv''_R\left(\frac{L}{2}\right)$$

$$\textcircled{8} EIv'''_L\left(\frac{L}{2}\right) = EIv'''_R\left(\frac{L}{2}\right)$$

Reorganize. Substituting the polynomials:

$$\textcircled{1} C_{4L} = 0$$

$$\textcircled{2} C_{2L} = 0$$

$$\textcircled{3} \frac{C_{1R}}{6}L^3 + \frac{C_{2R}}{2}L^2 + C_{3R}L + C_{4R} = 0$$

$$\textcircled{4} C_{1R}L + C_{2R} = 0$$

$$\textcircled{7} -\frac{w}{2}\frac{L^2}{4} + C_{1L}\frac{L}{2} = C_{1R}\frac{L}{2} + C_{2R}$$

$$\textcircled{8} -w\frac{L}{2} + C_{1L} = C_{1R}$$

$$\textcircled{5} -\frac{w}{24}\frac{L^4}{16} + \frac{C_{1L}}{6}\frac{L^3}{8} + C_{3L}\frac{L}{2} = \frac{C_{1R}}{6}\frac{L^3}{8} + \frac{C_{2R}}{2}\frac{L^2}{4} + C_{3R}\frac{L}{2} + C_{4R}$$

$$\textcircled{6} -\frac{w}{6}\frac{L^3}{8} + \frac{C_{1L}}{2}\frac{L^2}{4} + C_{3L} = \frac{C_{1R}}{2}\frac{L^2}{4} + C_{2R}\frac{L}{2} + C_{3R}$$

Solve the system. From ① and ②: $C_{4L} = 0$, $C_{2L} = 0$. The remaining six equations are solved by substitution (or any linear-solver): combining ⑦, ⑧ with ④ gives the right-region force constants, then ③, ⑤, ⑥ give the slope/deflection constants:

$$C_{1L} = \frac{3wL}{8}$$

$$C_{2L} = 0$$

$$C_{3L} = -\frac{9wL^3}{384}$$

$$C_{4L} = 0$$

$$C_{1R} = -\frac{wL}{8}$$

$$C_{2R} = \frac{wL^2}{8}$$

$$C_{3R} = -\frac{17wL^3}{384}$$

$$C_{4R} = \frac{wL^4}{384}$$

Note $C_{1L} = R_A = \frac{3wL}{8}$ and $C_{1R} = V_R = -\frac{wL}{8} = -R_B$: the integration constants recover the statics.

Resulting deflection functions:

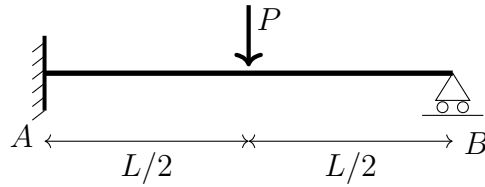
$$EIv_L(x) = -\frac{w}{24}x^4 + \frac{wL}{16}x^3 - \frac{9wL^3}{384}x, \quad 0 \leq x \leq \frac{L}{2},$$

$$EIv_R(x) = -\frac{wL}{48}x^3 + \frac{wL^2}{16}x^2 - \frac{17wL^3}{384}x + \frac{wL^4}{384}, \quad \frac{L}{2} \leq x \leq L.$$

End slopes and midspan deflection:

$$\theta_A = v'_L(0) = -\frac{9wL^3}{384EI} = -\frac{3wL^3}{128EI}, \quad \theta_B = v'_R(L) = \frac{7wL^3}{384EI}, \quad v\left(\frac{L}{2}\right) = -\frac{5wL^4}{768EI}.$$

The negative signs indicate a clockwise end rotation at A and downward deflection, as expected for downward load.

Example 2: propped cantilever with a point load at midspan

This beam is statically *indeterminate* (four support unknowns, three equilibrium equations), yet the double-integration procedure applies unchanged. No distributed load acts between the supports, so both regions are governed by the homogeneous equation:

$q(x) :$	$EIv_L''''(x) = 0$	$EIv_R''''(x) = 0$
$V(x) :$	$EIv_L'''(x) = C_{1L}$	$EIv_R'''(x) = C_{1R}$
$M(x) :$	$EIv_L''(x) = C_{1L}x + C_{2L}$	$EIv_R''(x) = C_{1R}x + C_{2R}$
slope:	$EIv_L'(x) = \frac{C_{1L}}{2}x^2 + C_{2L}x + C_{3L}$	$EIv_R'(x) = \frac{C_{1R}}{2}x^2 + C_{2R}x + C_{3R}$
deflection:	$EIv_L(x) = \frac{C_{1L}}{6}x^3 + \frac{C_{2L}}{2}x^2 + C_{3L}x + C_{4L}$	$EIv_R(x) = \frac{C_{1R}}{6}x^3 + \frac{C_{2R}}{2}x^2 + C_{3R}x + C_{4R}$

8 unknowns $\Rightarrow$ 8 equations. Boundary conditions – fixed support at A (deflection and slope), roller at B (deflection and moment):

$$\textcircled{1} v_L(0) = 0 \quad \textcircled{2} v'_L(0) = 0 \quad \textcircled{3} v_R(L) = 0 \quad \textcircled{4} EI v''_R(L) = 0$$

Continuity at $x = L/2$: deflection, slope, and moment are continuous, but the point load causes a *jump in shear*. Equilibrium of an infinitesimal element at the load (FBD: V_L up on the left face, P down, V_R down on the right face):

$$\textcircled{5} -V_L\left(\frac{L}{2}\right) + V_R\left(\frac{L}{2}\right) + P = 0 \implies -C_{1L} + C_{1R} = -P$$

$$\textcircled{6} EI v''_L\left(\frac{L}{2}\right) = EI v''_R\left(\frac{L}{2}\right) \quad \textcircled{7} v'_L\left(\frac{L}{2}\right) = v'_R\left(\frac{L}{2}\right) \quad \textcircled{8} v_L\left(\frac{L}{2}\right) = v_R\left(\frac{L}{2}\right)$$

Substitute the polynomial functions:

$$\textcircled{1} C_{4L} = 0$$

$$\textcircled{2} C_{3L} = 0$$

$$\textcircled{3} \frac{C_{1R}}{6} L^3 + \frac{C_{2R}}{2} L^2 + C_{3R} L + C_{4R} = 0$$

$$\textcircled{4} C_{1R} L + C_{2R} = 0$$

$$\textcircled{5} -C_{1L} + C_{1R} = -P$$

$$\textcircled{6} C_{1L} \frac{L}{2} + C_{2L} = C_{1R} \frac{L}{2} + C_{2R}$$

$$\textcircled{7} \frac{C_{1L}}{2} \frac{L^2}{4} + C_{2L} \frac{L}{2} + C_{3L} = \frac{C_{1R}}{2} \frac{L^2}{4} + C_{2R} \frac{L}{2} + C_{3R}$$

$$\textcircled{8} \frac{C_{1L}}{6} \frac{L^3}{8} + \frac{C_{2L}}{2} \frac{L^2}{4} + C_{3L} \frac{L}{2} + C_{4L} = \frac{C_{1R}}{6} \frac{L^3}{8} + \frac{C_{2R}}{2} \frac{L^2}{4} + C_{3R} \frac{L}{2} + C_{4R}$$

These eight linear equations are reorganized in matrix form on the next page and solved with Python (see the attached notebook output). The resulting constants are

$$C_{1L} = \frac{11P}{16}, \quad C_{2L} = -\frac{3PL}{16}, \quad C_{3L} = C_{4L} = 0, \quad C_{1R} = -\frac{5P}{16}, \quad C_{2R} = \frac{5PL}{16}, \quad C_{3R} = -\frac{PL^2}{8}, \quad C_{4R} = \frac{PL^3}{48},$$

giving $M_A = -\frac{3PL}{16}$ (hogging at the fixed end), prop reaction $R_B = \frac{5P}{16}$, and midspan deflection $v\left(\frac{L}{2}\right) = -\frac{7PL^3}{768EI}$.

Reorganize equations

$$\textcircled{1} 0 \times C_{1L} + 0 \times C_{2L} + 0 \times C_{3L} + 1 \times C_{4L} + 0 \times C_{1R} + 0 \times C_{2R} + 0 \times C_{3R} + 0 \times C_{4R} = 0$$

$$\textcircled{2} 0 \times C_{1L} + 0 \times C_{2L} + 0 \times C_{3L} + 1 \times C_{4L} + \frac{L^3}{6} \times C_{1R} + \frac{L^2}{2} \times C_{2R} + L \times C_{3R} + 0 \times C_{4R} = 0$$

$$\textcircled{3} 0 \times C_{1L} + 0 \times C_{2L} + 1 \times C_{3L} + 0 \times C_{4L} + 0 \times C_{1R} + 0 \times C_{2R} + 0 \times C_{3R} + 0 \times C_{4R} = 0$$

$$\textcircled{4} 0 \times C_{1L} + 0 \times C_{2L} + 0 \times C_{3L} + 0 \times C_{4L} + L \times C_{1R} + 1 \times C_{2R} + 0 \times C_{3R} + 0 \times C_{4R} = 0$$

$$\textcircled{5} -1 \times C_{1L} + 0 \times C_{2L} + 0 \times C_{3L} + 0 \times C_{4L} + 1 \times C_{1R} + 0 \times C_{2R} + 0 \times C_{3R} + 0 \times C_{4R} = -P$$

$$\textcircled{6} \frac{L}{2} \times C_{1L} + 1 \times C_{2L} + 0 \times C_{3L} + 0 \times C_{4L} + \frac{L}{2} \times C_{1R} + 1 \times C_{2R} + 0 \times C_{3R} + 0 \times C_{4R} = 0$$

$$\textcircled{7} \frac{L^2}{8} \times C_{1L} + \frac{L}{2} \times C_{2L} + 1 \times C_{3L} + 0 \times C_{4L} - \frac{L^2}{8} \times C_{1R} - \frac{L}{2} \times C_{2R} - 1 \times C_{3R} + 0 \times C_{4R} = 0$$

$$\textcircled{8} \frac{L^3}{48} \times C_{1L} + \frac{L^2}{8} \times C_{2L} + \frac{L}{2} \times C_{3L} + 1 \times C_{4L} - \frac{L^3}{48} \times C_{1R} - \frac{L^2}{8} \times C_{2R} - \frac{L}{2} \times C_{3R} - 1 \times C_{4R} = 0$$

In matrix form:

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{L^3}{6} & \frac{L^2}{2} & L & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & L & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{L}{2} & 1 & 0 & 0 & \frac{L}{2} & 1 & 0 & 0 \\ \frac{L^2}{8} & \frac{L}{2} & 1 & 0 & -\frac{L^2}{8} & -\frac{L}{2} & -1 & 0 \\ \frac{L^3}{48} & \frac{L^2}{8} & \frac{L}{2} & 1 & -\frac{L^3}{48} & -\frac{L^2}{8} & -\frac{L}{2} & -1 \end{bmatrix} \begin{bmatrix} C_{1L} \\ C_{2L} \\ C_{3L} \\ C_{4L} \\ C_{1R} \\ C_{2R} \\ C_{3R} \\ C_{4R} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -P \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

So, we can solve for the coefficients C_{iL} and C_{iR} using the inverse of the matrix.

$$\begin{bmatrix} C_{1L} \\ C_{2L} \\ C_{3L} \\ C_{4L} \\ C_{1R} \\ C_{2R} \\ C_{3R} \\ C_{4R} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{L^3}{6} & \frac{L^2}{2} & L & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & L & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{L}{2} & 1 & 0 & 0 & \frac{L}{2} & 1 & 0 & 0 \\ \frac{L^2}{8} & \frac{L}{2} & 1 & 0 & -\frac{L^2}{8} & -\frac{L}{2} & -1 & 0 \\ \frac{L^3}{48} & \frac{L^2}{8} & \frac{L}{2} & 1 & -\frac{L^3}{48} & -\frac{L^2}{8} & -\frac{L}{2} & -1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -P \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

lecture_note_1.4.2_beam_analysis_example_2

August 25, 2026

```
[1]: import sympy
from matplotlib import pyplot
import numpy

L = sympy.symbols('L')
P = sympy.symbols('P')
x = sympy.symbols('x')

matrix_A = sympy.Matrix([
    [0, 0, 0, 1, 0, 0, 0, 0],
    [0, 0, 0, 0, L**3/6, L**2/2, L, 1],
    [0, 0, 1, 0, 0, 0, 0, 0],
    [0, 0, 0, 0, L, 1, 0, 0],
    [-1, 0, 0, 0, 1, 0, 0, 0],
    [L/2, 1, 0, 0, -L/2, -1, 0, 0],
    [L**2/8, L/2, 1, 0, -L**2/8, -L/2, -1, 0],
    [L**3/48, L**2/8, L/2, 1, -L**3/48, -L**2/8, -L/2, -1]])

matrix_A
```

```
[1]: 
$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{L^3}{6} & \frac{L^2}{2} & L & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & L & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{L}{2} & 1 & 0 & 0 & -\frac{L}{2} & -1 & 0 & 0 \\ \frac{L^2}{8} & \frac{L}{2} & 1 & 0 & -\frac{L^2}{8} & -\frac{L}{2} & -1 & 0 \\ \frac{L^3}{48} & \frac{L^2}{8} & \frac{L}{2} & 1 & -\frac{L^3}{48} & -\frac{L^2}{8} & -\frac{L}{2} & -1 \end{bmatrix}$$

```

```
[ ]: # Beam configuration -- idealized support symbols (Table 2.1 convention)
# Fixed support at x=0, roller at x=L, point load P at x=L/2.

def draw_fixed_support(ax, x, y, size=0.06, side='left'):
    """Fixed support: end line with hatch marks (3 unknowns)."""
    s = -1 if side == 'left' else 1
    ax.plot([x, x], [y - size, y + size], 'k-', lw=2)
    for yy in numpy.linspace(y - size, y + size, 6):
        ax.plot([x, x + s*0.5*size], [yy, yy - 0.5*size], 'k-', lw=1)
```

```

def draw_pin_support(ax, x, y, size=0.06):
    """Pin support: triangle, ground line with hatches (2 unknowns)."""
    ax.plot([x, x - size, x + size, x], [y, y - size, y - size, y], 'k-', lw=1.5)
    ax.plot([x - 1.6*size, x + 1.6*size], [y - size, y - size], 'k-', lw=1.5)
    for xx in numpy.linspace(x - 1.4*size, x + 1.4*size, 5):
        ax.plot([xx, xx - 0.5*size], [y - size, y - 1.5*size], 'k-', lw=1)

def draw_roller_support(ax, x, y, size=0.06):
    """Roller support: triangle over circles, ground line (1 unknown)."""
    ax.plot([x, x - size, x + size, x], [y, y - size, y - size, y], 'k-', lw=1.5)
    r = 0.35 * size
    for xc in (x - 0.5*size, x + 0.5*size):
        ax.add_patch(pyplot.Circle((xc, y - size - r), r, fill=False, lw=1.2))
    ax.plot([x - 1.6*size, x + 1.6*size], [y - size - 2*r, y - size - 2*r], 'k-', lw=1.5)
    for xx in numpy.linspace(x - 1.4*size, x + 1.4*size, 5):
        ax.plot([xx, xx - 0.5*size], [y - size - 2*r, y - size - 2*r - 0.5*size], 'k-', lw=1)

def draw_point_load(ax, x, y, size=0.18, label='$P$'):
    """Downward point load arrow."""
    ax.annotate('', xy=(x, y), xytext=(x, y + size),
                arrowprops=dict(arrowstyle='->', color='k', lw=1.8))
    ax.text(x + 0.02, y + size, label, fontsize=13)

# --- beam configuration for this example: fixed at 0, roller at L ---
fig, ax = pyplot.subplots(figsize=(8, 2.4))
ax.plot([0, 1], [0, 0], 'k-', lw=3) # beam
draw_fixed_support(ax, 0, 0, side='left')
draw_roller_support(ax, 1, 0)
draw_point_load(ax, 0.5, 0)
ax.text(0.25, -0.10, '$L/2$', ha='center', fontsize=12)
ax.text(0.75, -0.10, '$L/2$', ha='center', fontsize=12)
ax.plot([0, 0], [-0.07, -0.13], 'k-', lw=0.5); ax.plot([0.5, 0.5], [-0.07, -0.13], 'k-', lw=0.5)
ax.plot([1, 1], [-0.07, -0.13], 'k-', lw=0.5)
ax.set_xlim(-0.15, 1.15); ax.set_ylim(-0.3, 0.35)
ax.set_aspect('equal'); ax.axis('off')
ax.set_title('Beam configuration: fixed-roller, point load $P$ at midspan')
pyplot.show()

```

```
[2]: vector_b = sympy.Matrix([0, 0, 0, 0, -P, 0, 0, 0])
```

```
[3]: vector_coefficients = matrix_A.inv() * vector_b
      vector_coefficients.T
```

```
[3]:  $\begin{bmatrix} \frac{11P}{16} & -\frac{3LP}{16} & 0 & 0 & -\frac{5P}{16} & \frac{5LP}{16} & -\frac{L^2P}{8} & \frac{L^3P}{48} \end{bmatrix}$ 
```

```
[4]: vector_coefficients_L = sympy.Matrix(vector_coefficients[:4])
      vector_coefficients_R = sympy.Matrix(vector_coefficients[4:])
```

```
[5]: function_v_L = vector_coefficients_L.T * sympy.Matrix([x**3/6, x**2/2, x, 1])
      function_slope_L = function_v_L.diff(x)
      function_moment_L = function_slope_L.diff(x)
      function_shear_L = function_moment_L.diff(x)

      function_v_R = vector_coefficients_R.T * sympy.Matrix([x**3/6, x**2/2, x, 1])
      function_slope_R = function_v_R.diff(x)
      function_moment_R = function_slope_R.diff(x)
      function_shear_R = function_moment_R.diff(x)
```

```
[6]: print(function_moment_L.subs(x, 0/2))
      print(function_moment_L.subs(x, L/2))
      print(function_moment_R.subs(x, L/2))
```

```
Matrix([[ -3*L*P/16]])
```

```
Matrix([[5*L*P/32]])
```

```
Matrix([[5*L*P/32]])
```

```
[9]: # Define beam length and load for plotting (using normalized values)
      L_val = 1.0 # Normalized beam length
      P_val = 1.0 # Normalized load

      # Create x values for left and right sections
      x_left = numpy.linspace(0, L_val/2, 100)
      x_right = numpy.linspace(L_val/2, L_val, 100)

      # Calculate moment values for each section
      moment_left = []
      for x_val in x_left:
          M = function_moment_L[0].subs([(x, x_val), (L, L_val), (P, P_val)])
          moment_left.append(float(M))

      moment_right = []
      for x_val in x_right:
          M = function_moment_R[0].subs([(x, x_val), (L, L_val), (P, P_val)])
          moment_right.append(float(M))

      # Calculate shear force values for each section
      shear_left = []
      for x_val in x_left:
```

```

    V = function_shear_L[0].subs([(x, x_val), (L, L_val), (P, P_val)])
    shear_left.append(float(V))

shear_right = []
for x_val in x_right:
    V = function_shear_R[0].subs([(x, x_val), (L, L_val), (P, P_val)])
    shear_right.append(float(V))

# Create figure with two subplots
fig, (ax1, ax2) = pyplot.subplots(2, 1, figsize=(10, 8))

# Plot moment diagram
ax1.plot(x_left, moment_left, 'b-', linewidth=2, label='Left section')
ax1.plot(x_right, moment_right, 'r-', linewidth=2, label='Right section')
ax1.axhline(y=0, color='k', linestyle='-', linewidth=0.5)
ax1.axvline(x=L_val/2, color='g', linestyle='--', alpha=0.5, label='Point load at L/2')
ax1.fill_between(x_left, 0, moment_left, alpha=0.3, color='blue')
ax1.fill_between(x_right, 0, moment_right, alpha=0.3, color='red')
ax1.set_xlabel('Position along beam (x/L)')
ax1.set_ylabel('Bending Moment (M/PL)')
ax1.set_title('Bending Moment Diagram')
ax1.grid(True, alpha=0.3)
ax1.legend()

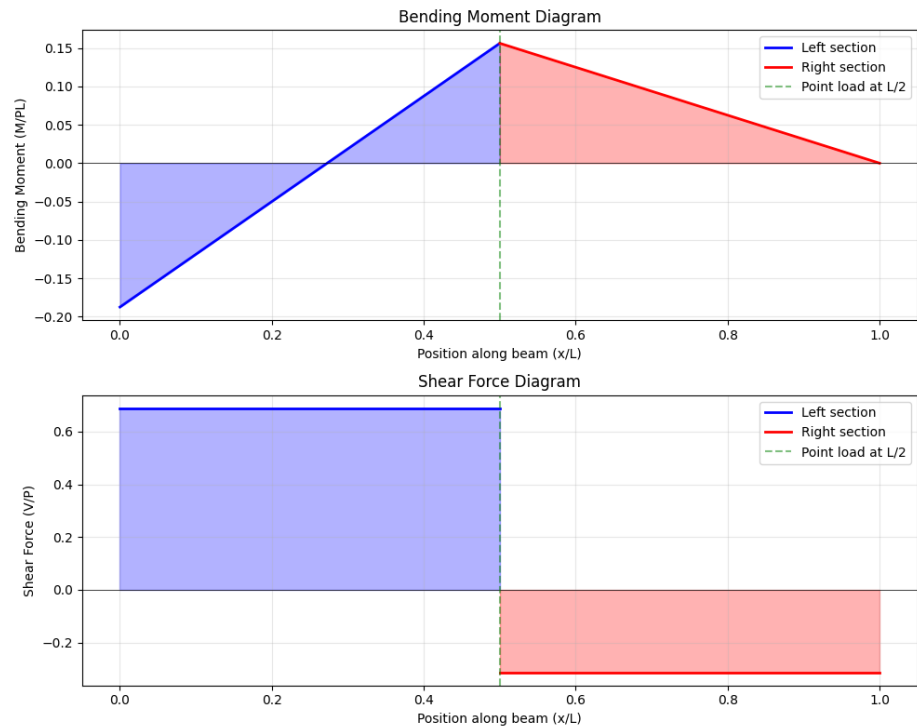
# Plot shear force diagram
ax2.plot(x_left, shear_left, 'b-', linewidth=2, label='Left section')
ax2.plot(x_right, shear_right, 'r-', linewidth=2, label='Right section')
ax2.axhline(y=0, color='k', linestyle='-', linewidth=0.5)
ax2.axvline(x=L_val/2, color='g', linestyle='--', alpha=0.5, label='Point load at L/2')
ax2.fill_between(x_left, 0, shear_left, alpha=0.3, color='blue')
ax2.fill_between(x_right, 0, shear_right, alpha=0.3, color='red')
ax2.set_xlabel('Position along beam (x/L)')
ax2.set_ylabel('Shear Force (V/P)')
ax2.set_title('Shear Force Diagram')
ax2.grid(True, alpha=0.3)
ax2.legend()

pyplot.tight_layout()
pyplot.show()

# Print key values
print("Key Values:")
print(f"Maximum moment at x=0: {moment_left[0]:.4f} PL")
print(f"Moment at x=L/2: {moment_left[-1]:.4f} PL")
print(f"Shear force at x=0 : {shear_left[0]:.4f} P")

```

```
print(f"Shear force at x=L/2 : {shear_right[0]:.4f} P")
print(f"Shear force at x=L: {shear_right[-1]:.4f} P")
```



Key Values:

Maximum moment at $x=0$: $-0.1875 PL$
 Moment at $x=L/2$: $0.1562 PL$
 Shear force at $x=0$: $0.6875 P$
 Shear force at $x=L/2$: $-0.3125 P$
 Shear force at $x=L$: $-0.3125 P$